

COMP6237 Data Mining

Lecture 6: Document Filtering

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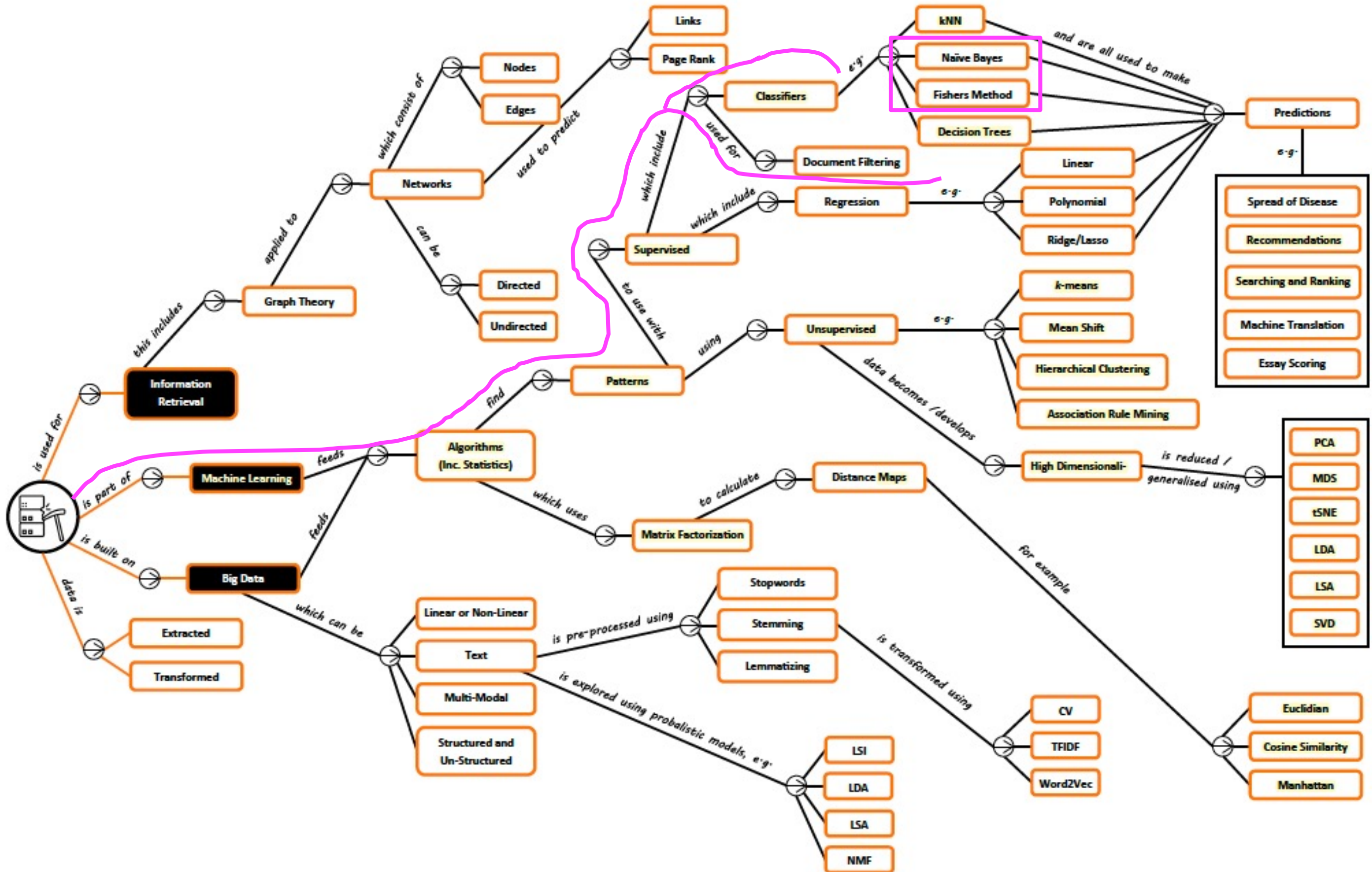
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Lecture slides available here:

<http://comp6237.ecs.soton.ac.uk/zh.html>

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Document Filtering – Roadmap



Document Filtering – Textbook

CHAPTER 6

Document Filtering

This chapter will demonstrate how to classify documents based on their contents, a very practical application of machine intelligence and one that is becoming more widespread. Perhaps the most useful and well-known application of document filtering is the elimination of spam. A big problem with the wide availability of email and the extremely low cost of sending email messages is that anyone whose address gets into the wrong hands is likely to receive unsolicited commercial email messages, making it difficult for them to read the messages that are actually of interest.

- ▶ **Programming Collective Intelligence: Building Smart Web 2.0 Applications**
T. Segaran.

Document Filtering – Overview (1/4)

Problem statement: Spam Filter

Need to classify if email is *spam* or *ham*

Early attempts used rules:

i.e. it is spam if it has:

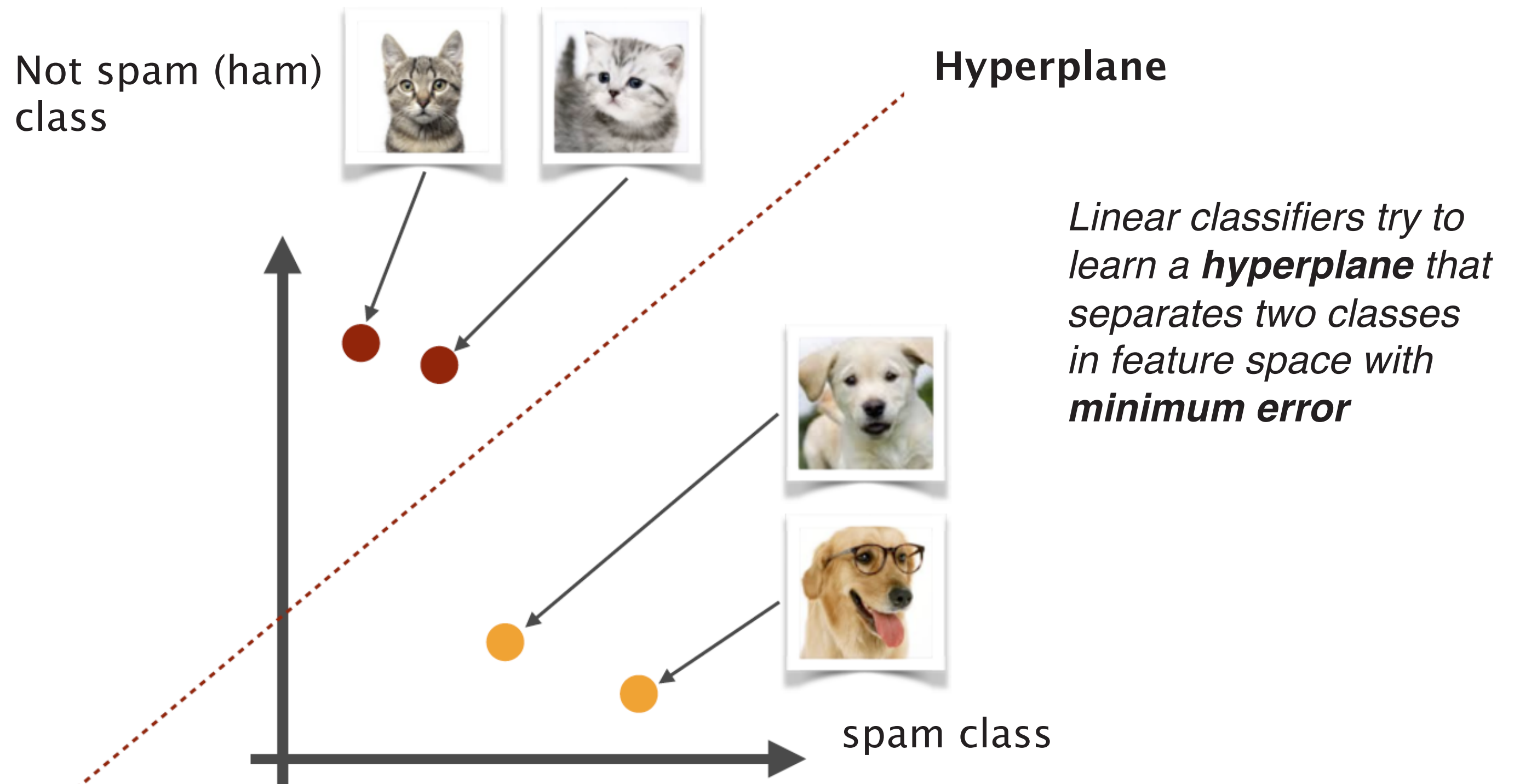
- ▶ Too many capitals
- ▶ The word viagra or penis
- ▶ Garish colors in embedded html
- ▶ Email only has an embedded picture

Problems..?

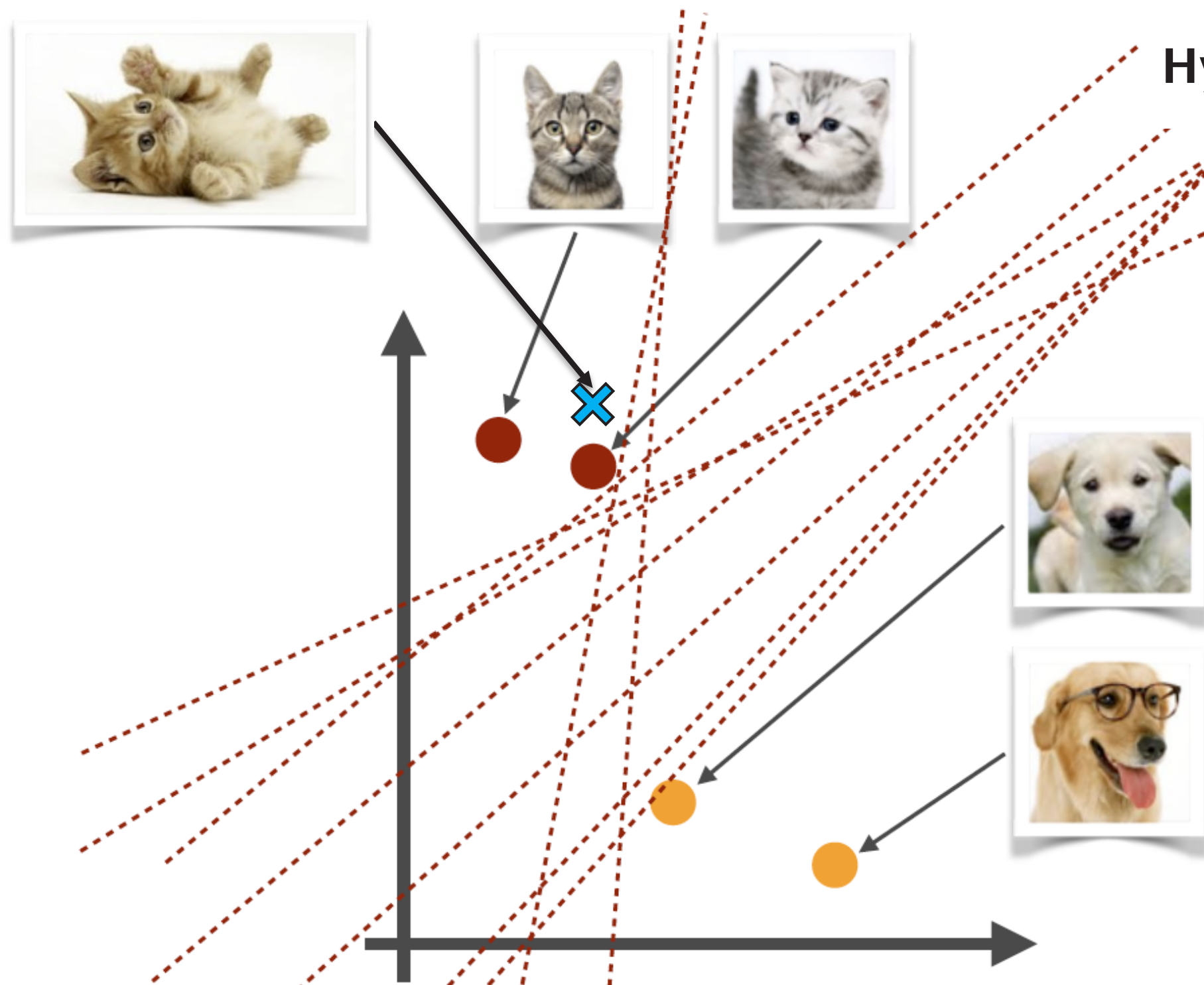
- ▶ Some people like to use uppercase
- ▶ Spammers learn rules and get round them
- ▶ No personalisation
- ▶ What if the user wants to buy viagra?

Solution: use Machine Learning to build personalised spam filters

Document Filtering – Overview (2/4)

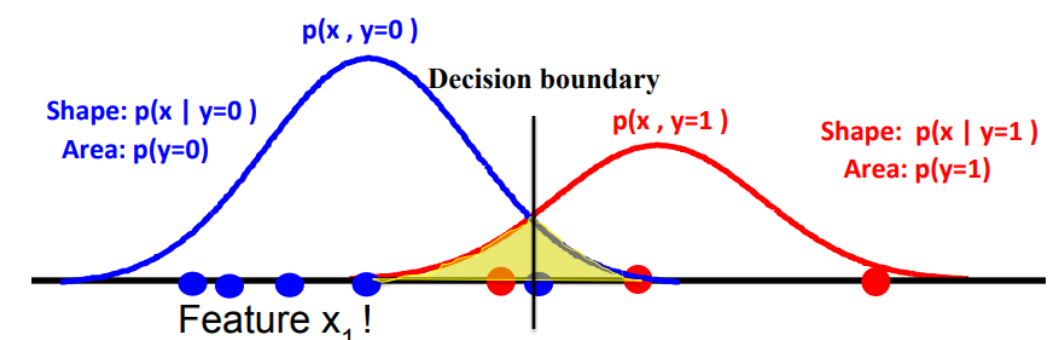


Document Filtering – Overview (3/4)



Hyperplanes

*Lots of hyperplanes
to choose from...
different linear
classification algorithms
apply differing
constraints when
learning the classifier*



Naïve Bayes

Document Filtering – Overview (4/4)



1. Priors

$$P(a) = \frac{4}{4+12} = 0.25 ; P(c) = 0.75$$

2. Likelihoods

$$p(h_x|c) = \frac{1}{\sqrt{2\pi}\sigma_{h,c}^2} \exp - \frac{1}{2} \left(\frac{(h_x - \mu_{h,c})^2}{\sigma_{h,c}^2} \right)$$

$$p(w_x|c) = \frac{1}{\sqrt{2\pi}\sigma_{w,c}^2} \exp - \frac{1}{2} \left(\frac{(w_x - \mu_{w,c})^2}{\sigma_{w,c}^2} \right)$$

$$p(h_x|a) = \frac{1}{\sqrt{2\pi}\sigma_{h,a}^2} \exp - \frac{1}{2} \left(\frac{(h_x - \mu_{h,a})^2}{\sigma_{h,a}^2} \right)$$

$$p(w_x|a) = \frac{1}{\sqrt{2\pi}\sigma_{w,a}^2} \exp - \frac{1}{2} \left(\frac{(w_x - \mu_{w,a})^2}{\sigma_{w,a}^2} \right)$$

Feature independence

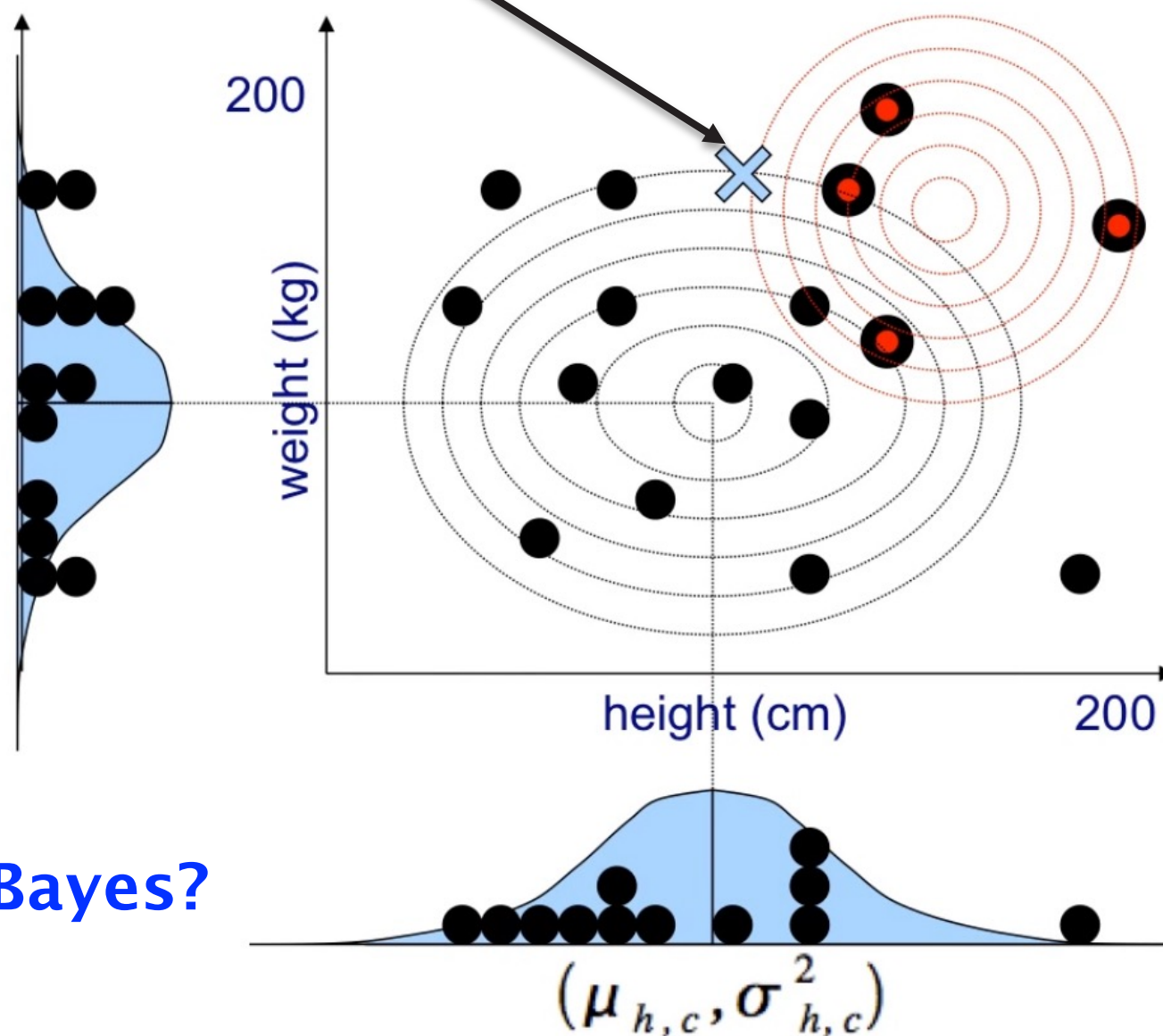
$$P(x|a) = p(h_x|a) p(w_x|a)$$

$$P(x|c) = p(h_x|c) p(w_x|c)$$

3. Posteriors

$$P(a|x) = \frac{P(x|a)P(a)}{P(x|a)P(a) + P(x|c)P(c)}$$

$$p(c|x) = \frac{p(x|c)p(c)}{p(x|a)p(a) + p(x|c)p(c)}$$



Naïve Bayes?

Fisher method? $P(a|x) = P(a|h_x)P(a|h_y)?$

Document Filtering – Learning Outcomes

- **LO1:** Demonstrate an understanding of the fundamental concepts and approaches for document filtering, such as: (exam)
 - ❖ Understanding the basic ideas behind Naive Bayes and Fisher's methods
 - ❖ Applying Naïve Bayes and Fisher's Methods to classify documents as spam or not spam
 - ❖ Discussing the pros and cons of using Naive Bayes vs. Fisher's methods for document filtering
- **LO2:** Implement the learned algorithms for document filtering (course work)

Assessment hints: Multi-choice Questions (single answer: concepts, calculation etc)

- *Textbook Exercises: textbooks (Programming + Mining)*
- *Other Exercises: <https://www-users.cse.umn.edu/~kumar001/dmbook/sol.pdf>*
- *ChatGPT or other AI-based techs*

Document Filtering – Naïve Bayes

The **Naïve** assumption allows the conditional probability to be expressed as the product of all the conditional feature probabilities

$$p(d|c) = \prod_{f \in d} p(f|c)$$

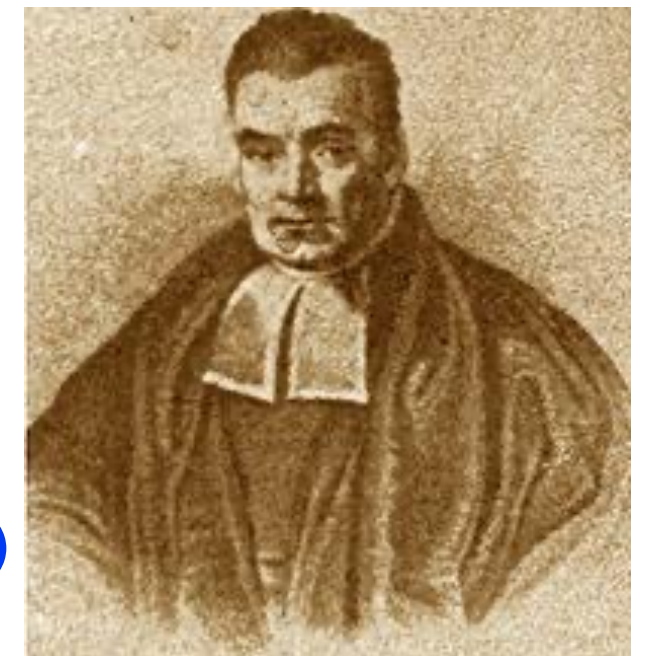
This is the **likelihood** of the document given a category.

Naive Bayes rules

$$\text{Posterior} \propto \text{Likelihood} \text{ Prior}$$

$$P(c|d) = \frac{P(d|c) P(c)}{p(d)}$$

- $P(c|d)$ = conditional probability of c given d (Posterior)
- $P(d|c)$ = conditional probability of d given c (Likelihood)
- $P(c)$ = prior probability of hypothesis/class c (Prior)
- $P(d)$ = prior probability of data d (Evidence)



Thomas Bayes (1702–1761)

Document Filtering – Naïve Bayes

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Naive Bayes rules

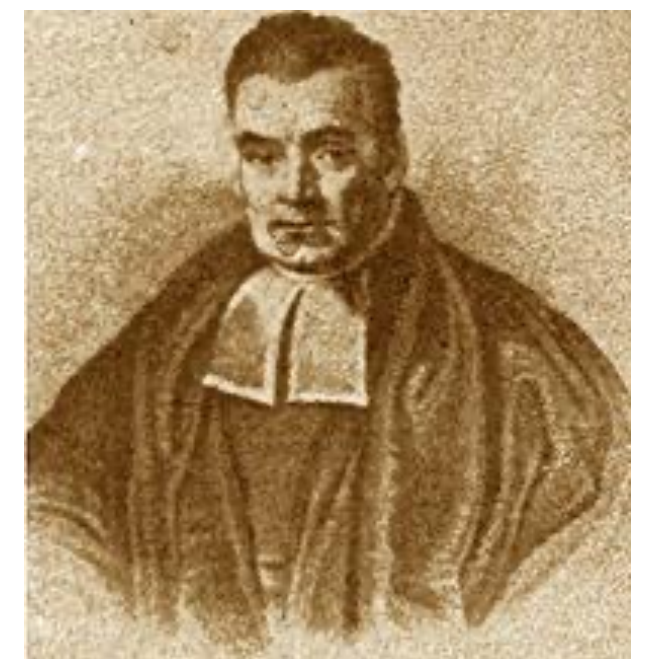
$$\text{Posterior} \propto \text{Likelihood} \text{ Prior}$$

$$P(c|d) = \frac{P(d|c) P(c)}{p(d)}$$

The **prior** can be calculated:

- ▶ Empirically, using the total No. docs in c divided by the total number
- ▶ or assuming all classes to be equally probably

$p(d)$ is constant, so therefore irrelevant, as same for all categories



Thomas Bayes (1702–1761)

Document Filtering – Naïve Bayes

Start with *bag of words*

- ▶ simple tokenisation: split on non-letters
- ▶ simple pre-processing: convert all to lower case
- ▶ count only presence or absence of term

Make a table counting the number of times a term has appeared for each category

	money	viagra	tea	you	dear	...	Total
Spam	15	8	0	19	15	...	30
Ham	2	0	15	20	12	...	70

Count up the total number of documents

Document Filtering – Naïve Bayes

Now work out the conditional probability, i.e. the probability of a feature given a category:

If n_c is the number of documents in a category, and n_{fc} is the number of documents with feature f in category c ..

Likelihood

$$p(f|c) = \frac{n_{fc}}{n_c}$$

Gaussian version

$$p(h_x|c) = \frac{1}{\sqrt{2\pi\sigma_{h,c}^2}} \exp - \frac{1}{2} \left(\frac{(h_x - \mu_{h,c})^2}{\sigma_{h,c}^2} \right)$$

	money	viagra	tea	you	dear	...	Total
Spam	15	8	0	19	15	...	30
Ham	2	0	15	20	12	...	70

Using the data above:

calculate $p(\text{"viagra"} | \text{Spam})$

Document Filtering – Naïve Bayes

Now work out the conditional probability, i.e. the probability of a feature given a category:

If n_c is the number of documents in a category, and n_{fc} is the number of documents with feature f in category c ..

Likelihood
$$p(f|c) = \frac{n_{fc}}{n_c}$$

	money	viagra	tea	you	dear	...	Total
Spam	15	8	0	19	15	...	30
Ham	2	0	15	20	12	...	70

Using the data above:

calculate $p(\text{"viagra"} | \text{Spam}) = 8/30 \approx 0.27$

calculate $p(\text{"tea"} | \text{Ham})$

Document Filtering – Naïve Bayes

Now work out the conditional probability, i.e. the probability of a feature given a category:

If n_c is the number of documents in a category, and n_{fc} is the number of documents with feature f in category c ..

Likelihood $p(f|c) = \frac{n_{fc}}{n_c}$

	money	viagra	tea	you	dear	...	Total
Spam	15	8	0	19	15	...	30
Ham	2	0	15	20	12	...	70

Using the data above:

calculate $p(\text{"viagra"} | \text{Spam}) = 8/30 \approx 0.27$

calculate $p(\text{"tea"} | \text{Ham}) = 15/70 \approx 0.21..$ but..

calculate $p(\text{"tea"} | \text{Spam})$

Document Filtering – Naïve Bayes

Now work out the conditional probability, i.e. the probability of a feature given a category:

If n_c is the number of documents in a category, and n_{fc} is the number of documents with feature f in category c ..

$$p(f|c) = \frac{n_{fc}}{n_c}$$

	money	viagra	tea	you	dear	...	Total
Spam	15	8	0	19	15	...	30
Ham	2	0	15	20	12	...	70

Using the data above:

calculate $p(\text{"viagra"}|Spam) = 8/30 \approx 0.27$

calculate $p(\text{"tea"}|Ham) = 15/70 \approx 0.21..$ but..

calculate $p(\text{"tea"}|Spam) = 0/30 = 0 !..$ does this make sense?

Document Filtering – Naïve Bayes

Smoothing probability estimates

Should “tea” *never* be expected to appear in Spam documents?

Need a better way to estimate the conditional probability that accounts for infrequently seen features (sample size too small)

We introduce an *assumed* probability

- ▶ where there is little evidence
- ▶ can be based on some evidence

Produce a weighted estimate for the conditional probability based on the assumed and the raw computed probability

Document Filtering – Naïve Bayes

Using Bayesian Statistics, assuming that random variables are drawn from probability distributions rather than point samples

In this case, Spam/Ham classification for a feature f is binomial with a *beta* distributed **prior**

$$p_w(f|c) = \frac{\textit{weight} \times \textit{assumed} + \textit{count} \times p_{raw}(f|c)}{\textit{count} + \textit{weight}}$$

where *count* is the number of times feature f occurs across all categories, *weight* is the strength we will give that assumption

In practice, the values for ***weight*** and ***assumed*** are found through testing to optimize performance. Reasonable starting points are 1 and .5 for *weight* and *assumed*, respectively

Document Filtering – Naïve Bayes

However the conditional probability of the whole document is required

To do this we assume that all features are independent of each other.

This is what makes it **Naïve** Bayes

In this case the *features* are words, and as certain words are very likely to appear together this assumption is false. However in practice, it doesn't matter.

It will still work even if incorrect!

Document Filtering – Naïve Bayes

The **Naïve** assumption allows the conditional probability to be expressed as the product of all the conditional feature probabilities

$$p(d|c) = \prod_{f \in d} p(f|c)$$

This is the **likelihood** of the document given a category.

When implementing, the product of a lot of very small numbers could lead to *floating point underflow*, so we take the logs and sum instead.

$$\log(p(c|d)) \propto \log(p(c)) + \sum_{f \in d} \log(p(f|c))$$

$$P(c|d) = \frac{P(d|c) P(c)}{p(d)}$$

Document Filtering – Naïve Bayes

posterior $\propto$ likelihood $\times$ prior

$$\therefore p(c|d) \propto p(d|c) \times p(c)$$

$p(c|d)$ can be calculated for all categories using this formula

The most likely category is thus assigned to the document, i.e. with the largest $p(c|d)$. This is **maximum a posteriori**, MAP, and assumes the categories are equal.

In the case of Spam/Ham this isn't true, the cost of misclassifying a good email as Spam is much higher than misclassifying a Spam email as Ham.

Document Filtering – Naïve Bayes

Could require a higher ratio to classify..

$$\text{if } \frac{p(\text{Spam}|d)}{p(\text{Ham}|d)} > 3 \quad \text{then } \text{Spam}$$

Document Filtering – Fisher's Method vs. Naïve Bayes

Naïve Bayes: uses feature likelihoods to compute whole document probability

$$\log(p(c|d)) \propto \log(p(c)) + \sum_{f \in d} \log(p(f|c))$$

Fisher's Method

- ▶ calculates probability of each category for each feature $p(c|f)$
- ▶ tests to see if combined probabilities are more or less likely than a random

This assumes independence of features

$$p(c|d) = K^{-1}\left(-2 \sum_{i=1}^k \ln(p(c = C|f_i)), 2k\right) = K^{-1}\left(-2 \ln\left(\prod_{i=1}^k p(c = C|f_i)\right), 2k\right)$$

where K^{-1} is the inverse chi-squared function

Document Filtering – Fisher's Method

To calculate category probabilities $p(c|f)$ for given feature:
Can use Bayes' Extended form, but need $P(c)$ for all c

$$p(c|f) = \frac{p(f|c_i)p(c_i)}{\sum_j p(f|c_j)p(c_j)}$$

$p(c_j)$ can be estimated from the data, or from an unbiased estimate,
all $p(c_j)$ equally likely

Document Filtering – Fisher's Method

$$p(c|f) = \frac{p(f|c_i)p(c_i)}{\sum_j p(f|c_j)p(c_j)}$$

For a Spam/Ham classifier, there is no *a priori* reason to assume one or the other.

$$P(c) = P(Ham) = P(Spam) = 0.5$$

$$\therefore P(c|f) = \frac{P(f|c)}{(P(f|Ham) + P(f|Spam))}$$

Document Filtering – Fisher's Method

Fisher's method combines k probabilities ("p-values") $P(c = C|f_k)$ from each test in to a single test statistic, X^2

$$X_{2k}^2 \sim -2 \sum \ln(p(c = C|f_i))$$

if the p-values are independent, this X^2 statistic will follow a *chi-squared* distribution in $2k$ degrees of freedom

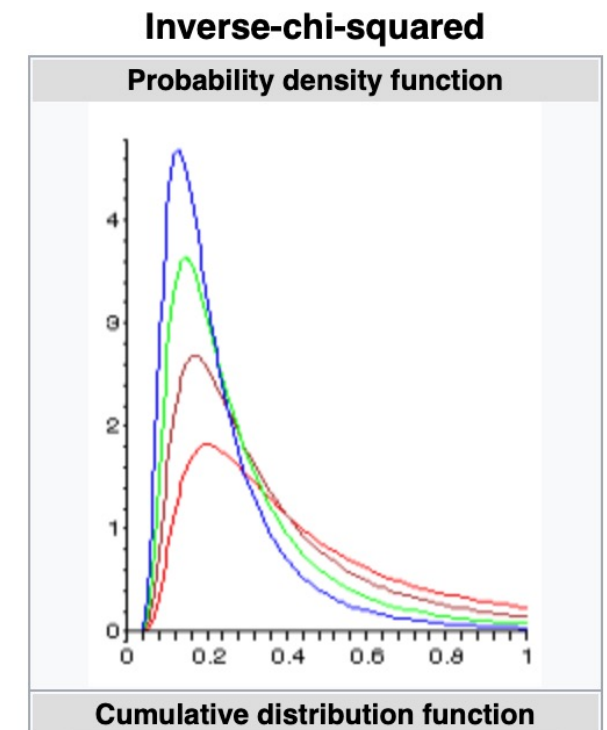
So we can calculate a combined p-value

$$p_C = K^{-1}\left(-2 \sum_{i=1}^k \ln(p(c = C|f_i)), 2k\right) = K^{-1}\left(-2 \ln\left(\prod_{i=1}^k p(c = C|f_i)\right), 2k\right)$$

where K^{-1} is the inverse chi-squared function

PDF	$\frac{2^{-\nu/2}}{\Gamma(\nu/2)} x^{-\nu/2-1} e^{-1/(2x)}$	Parameters	$\nu > 0$
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<https://en.wikipedia.org>



<https://en.wikipedia.org>

Document Filtering – Fisher's Method

Making classifications

$$I = \frac{1 + p_{Spam} - p_{Ham}}{2}$$

I tends to 1 if document is Spam, and 0 if it is Ham.

Document Filtering – Summary

Learning models to categorise data is a core part of data mining

- ▶ Many supervised machine learning techniques can be used
- ▶ For Spam/Ham, probabilistic approaches work well as:
 - ▶ easy to implement
 - ▶ interpretable
 - ▶ computationally efficient
 - ▶ online
 - ▶ .. but assuming independence of features (naïve) can be problematic

Choice of features is key

Appendix – Improving Text Features

Feature Engineering:

In email, there are *Fields*

- ▶ sender; to; cc;
- ▶ title; subject; body

These can be used as separate features

Appendix – Improving Text Features

Spam emails may contain lots of capitals, so a new feature can be engineered to measure this

e.g. if more than 30% of words are uppercase, then record a virtual 'uppercase' feature..

	money	viagra	tea	you	dear	...	v. upper	Total
Spam	15	8	0	19	15	...	5	30
Ham	2	0	15	20	12	...	0	70

Appendix – Improving Text Features

N-Grams

Bag of words loses the order and context the words are in, so instead of looking at just individual words, we can use pairs, or triplets ($n=2, 3$) of words as a base feature.

Generally these are called *n – grams*

"The quick brown fox jumped over the lazy dog"
becomes:

'The quick' "quick brown" "brown fox" "fox jumped" "jumped over"
"over the" "the lazy" "lazy dog"

Appendix – Improving Text Features

Can we do better?

Natural Language Processing

► POS - Part of Speech tagging

"The quick brown fox jumped over the lazy dog"

[('The', 'DT'), ('quick', 'JJ'), ('brown', 'JJ'),
('fox', 'NN'), ('jumped', 'VBD'), ('over', 'IN'),
('the', 'DT'), ('lazy', 'JJ'), ('dog', 'NN')]

"The quick brown fox jumped over the lazy dog"

ipynb demo

<https://github.com/zhiwu-huang/Data-Mining-Demo-Code-18-19>

Appendix – Improving Text Features

- ▶ NE - Named entity extraction

"but unfortunately for MSC Bellissima, the
British weather intervened."

but/CC

unfortunately/RB

for/IN

(ORGANIZATION MSC/NNP Bellissima/NNP)

the/DT

(GPE British/JJ)

weather/NN

intervened/VBD

Appendix – Improving Text Features

Using these tools we can get $\approx 91\%$ accuracy for POS tagging

State of the art is close to 97% accurate.

Natural Language Processing is **hard**!

More accurate, robust NLP is shallow
i.e. superficial analysis, not really understanding sentence

ChatGPT

Autoregressive Language Models

Simple, Old idea. Complete the sentence “*The mouse ate the*” ranked by probability learned from the corpus.

$$p(\text{“the”, “mouse”, “ate”, “the”, “cheese”}) = 0.10$$

$$p(\text{“the”, “mouse”, “ate”, “the”, “mouse”}) = 0.01$$

$$p(\text{“the”, “mouse”, “ate”, “the”, “house”}) = 0.0001$$

Neural Language Models.

- **Neural nets** “compactly” represent that probability function